

25.1-6

In a *d-regular* graph, every vertex has degree d . If $G = (V, E)$ is bipartite with vertex partition $V = L \cup R$ and also d -regular, then $|L| = |R|$. Use Hall's theorem (see Exercise 25.1-5) to prove that every d -regular bipartite graph contains a perfect matching. Then use that result to prove that every d -regular bipartite graph contains d disjoint perfect matchings.

25.2 The stable-marriage problem

In Section 25.1, the goal was to find a maximum matching in an undirected bipartite graph. If you know that the graph $G = (V, E)$ with vertex partition $V = L \cup R$ is a *complete bipartite graph*¹—containing an edge from every vertex in L to every vertex in R —then you can find a maximum matching by a simple greedy algorithm.

When a graph can have several matchings, you might want to decide which matchings are most desirable. In Section 25.3, we'll add weights to the edges and find a matching of maximum weight. In this section, we will instead add some information to each vertex in a complete bipartite graph: a ranking of the vertices in the other side. That is, each vertex in L has an ordered list of all the vertices in R , and vice-versa. To keep things simple, let's assume that L and R each contain n vertices. The goal here is to match each vertex in L with a vertex in R in a “stable” way.

This problem derives its name, the *stable-marriage problem*, from the notion of heterosexual marriage, viewing L as a set of women and R as a set of men.² Each woman ranks all the men in terms of desirability, and each man does the same with all the women. The goal is to pair up women and men (a matching) so that if a woman and a man are not matched to each other, then at least one of them prefers their assigned partner.

If a woman and a man are not matched to each other but each prefers the other over their assigned partner, they form a *blocking pair*. A blocking pair has incentive to opt out of the assigned pairing and get

together on their own. If that were to occur, then this pair would block the matching from being “stable.” A *stable matching*, therefore, is a matching that has no blocking pair. If there is a blocking pair, then the matching is *unstable*.

Let’s look at an example with four women—Wanda, Emma, Lacey, and Karen—and four men—Oscar, Davis, Brent, and Hank—having the following preferences:

Wanda: Brent, Hank, Oscar, Davis
Emma: Davis, Hank, Oscar, Brent
Lacey: Brent, Davis, Hank, Oscar
Karen: Brent, Hank, Davis, Oscar

Oscar: Wanda, Karen, Lacey, Emma
Davis: Wanda, Lacey, Karen, Emma
Brent: Lacey, Karen, Wanda, Emma
Hank: Lacey, Wanda, Emma, Karen

A stable matching comprises the following pairs:

Lacey and Brent
Wanda and Hank
Karen and Davis
Emma and Oscar

You can verify that this matching has no blocking pair. For example, even though Karen prefers Brent and Hank to her partner Davis, Brent prefers his partner Lacey to Karen, and Hank prefers his partner Wanda to Karen, so that neither Karen and Brent nor Karen and Hank form a blocking pair. In fact, this stable matching is unique. Suppose instead that the last two pairs were

Emma and Davis
Karen and Oscar

Then Karen and Davis would be a blocking pair, because they were not paired together, Karen prefers Davis to Oscar, and Davis prefers Karen to Emma. Therefore, this matching is not stable.

Stable matchings need not be unique. For example, suppose that there are three women—Monica, Phoebe, and Rachel—and three men—Chandler, Joey, and Ross—with these preferences:

Monica: Chandler, Joey, Ross
 Phoebe: Joey, Ross, Chandler
 Rachel: Ross, Chandler, Joey
 Chandler: Phoebe, Rachel, Monica
 Joey: Rachel, Monica, Phoebe
 Ross: Monica, Phoebe, Rachel

In this case, there are three stable matchings:

Matching 1	Matching 2	Matching 3
Monica and Chandler	Phoebe and Chandler	Rachel and Chandler
Phoebe and Joey	Rachel and Joey	Monica and Joey
Rachel and Ross	Monica and Ross	Phoebe and Ross

In matching 1, all women get their first choice and all men get their last choice. Matching 2 is the opposite, with all men getting their first choice and all women getting their last choice. When all the women or all the men get their first choice, there plainly cannot be a blocking pair. In matching 3, everyone gets their second choice. You can verify that there are no blocking pairs.

You might wonder whether it is always possible to come up with a stable matching no matter what rankings each participant provides. The answer is yes. (Exercise 25.2-3 asks you to show that even in the scenario of the National Resident Matching Program, where each hospital takes on multiple students, it is always possible to devise a stable assignment.) A simple algorithm known as the Gale-Shapley algorithm always finds a stable matching. The algorithm has two variants, which mirror each other: “woman-oriented” and “man-oriented.” Let’s examine the woman-oriented version. Each participant is either “free” or “engaged.” Everyone starts out free. Engagements occur when a free woman proposes to a man. When a man is first proposed to, he goes from free to engaged, and he always stays

engaged, though not necessarily to the same woman. If an engaged man receives a proposal from a woman whom he prefers to the woman he's currently engaged to, that engagement is broken, the woman to whom he had been engaged becomes free, and the man and the woman whom he prefers become engaged. Each woman proposes to the men in her preference list, in order, until the last time she becomes engaged. When a woman is engaged, she temporarily stops proposing, but if she becomes free again, she continues down her list. Once everyone is engaged, the algorithm terminates. The procedure GALE-SHAPLEY on the next page makes this process more concrete. The procedure allows for some choice: any free woman may be selected in line 2. We'll see that the procedure produces a stable matching regardless of the order in which line 2 chooses free women. For the man-oriented version, just reverse the roles of men and women in the procedure.

Let's see how the GALE-SHAPLEY procedure executes on the example with Wanda, Emma, Lacey, Karen, Oscar, Davis, Brent, and Hank. After everyone is initialized to free, here is one possible version of what can occur in successive iterations of the **while** loop of lines 2–9:

1. Wanda proposes to Brent. Brent is free, so that Wanda and Brent become engaged and no longer free.
2. Emma proposes to Davis. Davis is free, so that Emma and Davis become engaged and no longer free.
3. Lacey proposes to Brent. Brent is engaged to Wanda, but he prefers Lacey. Brent breaks the engagement to Wanda, who becomes free. Lacey and Brent become engaged, with Lacey no longer free.
4. Karen proposes to Brent. Brent is engaged to Lacey, whom he prefers to Karen. Brent rejects Karen, who remains free.
5. Karen proposes to Hank. Hank is free, so that Karen and Hank become engaged and no longer free.
6. Wanda proposes to Hank. Hank is engaged to Karen, but he prefers Wanda. Hank breaks the engagement with Karen, who

becomes free. Wanda and Hank become engaged, with Wanda no longer free.

7. Karen proposes to Davis. Davis is engaged to Emma, but he prefers Karen. Davis breaks the engagement to Emma, who becomes free. Karen and Davis become engaged, with Karen no longer free.
8. Emma proposes to Hank. Hank is engaged to Wanda, whom he prefers to Emma. Hank rejects Emma, who remains free.
9. Emma proposes to Oscar. Oscar is free, so that Emma and Oscar become engaged and no longer free.

GALE-SHAPLEY (*men, women, rankings*)

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1 assign each woman and man as free
2 while some woman  $w$  is free
3   let  $m$  be the first man on  $w$ 's ranked list to whom she has not
   proposed
4   if  $m$  is free
5      $w$  and  $m$  become engaged to each other (and not free)
6   elseif  $m$  ranks  $w$  higher than the woman  $w'$  he is currently
   engaged to
7      $m$  breaks the engagement to  $w'$ , who becomes free
8      $w$  and  $m$  become engaged to each other (and not free)
9   else  $m$  rejects  $w$ , with  $w$  remaining free
10 return the stable matching consisting of the engaged pairs
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At this point, everyone is engaged and nobody is free, so the **while** loop terminates. The procedure returns the stable matching we saw earlier.

The following theorem shows that not only does GALE-SHAPLEY terminate, but that it always returns a stable matching, thereby proving that a stable matching always exists.

Theorem 25.9

The procedure GALE-SHAPLEY always terminates and returns a stable matching.

Proof Let's first show that the **while** loop of lines 2–9 always terminates, so that the procedure terminates. The proof is by contradiction. If the loop fails to terminate, it is because some woman remains free. In order for a woman to remain free, she must have proposed to all the men and been rejected by each one. In order for a man to reject a woman, he must be already engaged. Therefore, all the men are engaged. Once engaged, a man stays engaged (though not necessarily to the same woman). There are an equal number n of women and men, however, which means that every woman is engaged, leading to the contradiction that no women are free. We must also show that the **while** loop makes a bounded number of iterations. Since each of the n women goes through her ranking of the n men in order, possibly not reaching the end of her list, the total number of iterations is at most n^2 . Therefore, the **while** loop always terminates, and the procedure returns a matching.

We need to show that there are no blocking pairs. We first observe that once a man m is engaged to a woman w , all subsequent actions for m occur in lines 6–8. Therefore, once a man is engaged, he stays engaged, and any time he breaks an engagement to a woman w , it's for a woman whom he prefers to w . Suppose that a woman w is matched with a man m , but she prefers man m' . We'll show that w and m' is not a blocking pair, because m' does not prefer w to his partner. Because w ranks m' higher than m , she must have proposed to m' before proposing to m , and m' either rejected her proposal or accepted it and later broke the engagement. If m' rejected the proposal from w , it is because he was already engaged to some woman he prefers to w . If m' accepted and later broke the engagement, he was at some point engaged to w but later accepted a proposal from a woman he prefers to w . In either case, he ultimately ends up with a partner whom he prefers to w . We conclude that even though w might prefer m' to her partner m , it is not also the case that m' prefers w to his partner. Therefore, the procedure returns a matching containing no blocking pairs.

■

Exercise 25.2-1 asks you to provide the proof of the following corollary.

Corollary 25.10

Given preference rankings for n women and n men, the Gale-Shapley algorithm can be implemented to run in $O(n^2)$ time.

■

Because line 2 can choose any free woman, you might wonder whether different choices can produce different stable matchings. The answer is no: as the following theorem shows, every execution of the GALE-SHAPLEY produces exactly the same result. Moreover, the stable matching returned is optimal for the women.

Theorem 25.11

Regardless of how women are chosen in line 2 of GALE-SHAPLEY, the procedure always returns the same stable matching, and in this stable matching, each woman has the best partner possible in any stable matching.

Proof The proof that each woman has the best partner possible in any stable matching is by contradiction. Suppose that the GALE-SHAPLEY procedure returns a stable matching M , but that there is a different stable matching M' in which some woman w prefers her partner m' to the partner m she has in M . Because w ranks m' higher than m , she must have proposed to m' before proposing to m . Then there is a woman w' whom m' prefers to w , and m' was already engaged to w' when w proposed or m' accepted the proposal from w and later broke the engagement in favor of w' . Either way, there is a moment when m' decided against w in favor of w' . Now suppose, without loss of generality, that this moment was the first time that any man rejected a partner who belongs to some stable matching.

We claim that w' cannot have a partner m'' in a stable matching whom she prefers to m' . If there were such a man m'' , then in order for w' to propose to m' , she would have proposed to m'' and been rejected at some point before proposing to m' . If m' accepted the proposal from w and later broke it to accept w' , then since this was the first rejection in a

stable matching, we get the contradiction that m'' could not have rejected w' beforehand. If m'' was already engaged to w' when w proposed, then again, m'' could not have rejected w' beforehand, thus proving the claim.

Since w' does not prefer anyone to m' in a stable matching and w' is not matched with m' in M' (because m' is matched with w in M'), w' prefers m' to her partner in M' . Since w' prefers m' over her partner in M' and m' prefers w' over his partner w in M' , the pair w' and m' is a blocking pair in M' . Because M' has a blocking pair, it cannot be a stable matching, thereby contradicting the assumption that there exists some stable matching in which each woman has the best partner possible other than the matching M returned by GALE-SHAPLEY.

We put no condition on the execution of the procedure, which means that all possible orders in which line 2 selects women result in the same stable matching being returned.

■

Corollary 25.12

There can be stable matchings that the GALE-SHAPLEY procedure does not return.

Proof Theorem 25.11 says that for a given set of rankings, GALE-SHAPLEY returns just one matching, no matter how it chooses women in line 2. The earlier example of three women and three men with three different stable matchings shows that there can be multiple stable matchings for a given set of rankings. A call of GALE-SHAPLEY is capable of returning only one of these stable matchings.

■

Although the GALE-SHAPLEY procedure gives the best possible outcome for the women, the following corollary shows that it also produces the worst possible outcome for the men.

Corollary 25.13

In the stable matching returned by the procedure GALE-SHAPLEY, each man has the worst partner possible in any stable matching.

Proof Let M be the matching returned by a call to GALE-SHAPLEY. Suppose that there is another stable matching M' and a man m who prefers his partner w in M to his partner w' in M' . Let the partner of w in M' be m' . By Theorem 25.11, m is the best partner that w can have in any stable matching, which means that w prefers m to m' . Since m prefers w to w' , the pair w and m is a blocking pair in M' , contradicting the assumption that M' is a stable matching. ■

Exercises

25.2-1

Describe how to implement the Gale-Shapley algorithm so that it runs in $O(n^2)$ time.

25.2-2

Is it possible to have an unstable matching with just two women and two men? If so, provide and justify an example. If not, argue why not.

25.2-3

The National Resident Matching Program differs from the scenario for the stable-marriage problem set out in this section in two ways. First, a hospital may be matched with more than one student, so that hospital h takes $r_h \geq 1$ students. Second, the number of students might not equal the number of hospitals. Describe how to modify the Gale-Shapley algorithm to fit the requirements of the National Resident Matching Program.

25.2-4

Prove the following property, which is known as *weak Pareto optimality*:

Let M be the stable matching produced by the GALE-SHAPLEY procedure, with women proposing to men. Then, for a given instance of the stable-marriage problem there is no matching—stable or unstable—such that every woman has a

partner whom she prefers to her partner in the stable matching M .

25.2-5

The *stable-roommates problem* is similar to the stable-marriage problem, except that the graph is a complete graph, not bipartite, with an even number of vertices. Each vertex represents a person, and each person ranks all the other people. The definitions of blocking pairs and stable matching extend in the natural way: a blocking pair comprises two people who both prefer each other to their current partner, and a matching is stable if there are no blocking pairs. For example, consider four people—Wendy, Xenia, Yolanda, and Zelda—with the following preference lists:

Wendy: Xenia, Yolanda, Zelda
Xenia: Wendy, Zelda, Yolanda
Yolanda: Wendy, Zelda, Xenia
Zelda: Xenia, Yolanda, Wendy

You can verify that the following matching is stable:

Wendy and Xenia
Yolanda and Zelda

Unlike the stable-marriage problem, the stable-roommates problem can have inputs for which no stable matching exists. Find such an input and explain why no stable matching exists.

25.3 The Hungarian algorithm for the assignment problem

Let us once again add some information to a complete bipartite graph $G = (V, E)$, where $V = L \cup R$. This time, instead of having the vertices of each side rank the vertices on the other side, we assign a weight to each edge. Again, let's assume that the vertex sets L and R each contain n vertices, so that the graph contains n^2 edges. For $l \in L$ and $r \in R$, denote the weight of edge (l, r) by $w(l, r)$, which represents the utility gained by matching vertex l with vertex r .